ferryGL: Token-Free Automation for Syncing Cross-Organization GitHub Work into Linear

One-way, least-privilege sync that people and AI agents can run without spending model tokens

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Abstract

AI agents can now do much of the routine work around software projects, but every run of an agent costs model tokens, and routine work recurs. This paper makes the case for *token-free automation*: deterministic programs that do a routine, rule-based task without running a language model, and that people and AI agents can call instead of doing the task themselves. ("Token-free" refers to model tokens; such a program still authenticates with access tokens.) It presents ferryGL, an open-source, token-free tool for one such task. People who contribute to GitHub repositories that belong to another organization, such as a client, a partner or an open-source project, often plan that work in their own team's Linear workspace, which Linear's GitHub integration cannot reach: it must be installed by an owner or administrator on the GitHub side, and a GitHub organization can be connected to only one Linear workspace. ferryGL copies the issues and pull requests a person is part of, one way, from GitHub into their own Linear workspace, using only that person's credentials. It has no server and no database: it keeps its state in the metadata of the GitHub links it attaches to Linear issues, and moves a Linear status only when the GitHub state recorded there changes, so a status set by hand stays. A pure planner separates every read from every write, which makes each run previewable and every rerun idempotent. The paper states eight safety invariants and evaluates them with a randomized, model-based simulation of 2,000 histories (74,139 runs, with 5 % of the writes of applying runs failing), in which every safety property held in all 413,071 of its checks; a metamorphic check found one ordering edge case, which the paper analyzes. A request-cost model and a planner benchmark complete the evaluation. Finally, measured workloads priced at published rates estimate what an AI agent would spend: doing the job itself, 91,620 to 217,857 tokens on a run of a 100-item route that finds 4 changes to make, and 54,966 to 106,048 on one that finds none; delegating it to ferryGL, 27 to 38 times fewer, and up to 934 times fewer on 800 items. Across 24 current models from 10 providers, a team of 100 people would spend an estimated $596 to $59,576 a year under the plausible assumptions on an agent doing the job itself twice a workday, $58.99 to $6,249 on one delegating it to ferryGL, and nothing on inference with ferryGL alone.

Keywords: token-free automation, token-free synchronization, AI agents, LLM agents, agent tool use, inference cost, deterministic automation, issue tracking, one-way synchronization, least privilege, idempotence, edge-triggered reconciliation, GitHub, Linear

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1 Introduction

AI agents that reason and call tools can now do much of the routine work around software projects [75, 76]. Routine work, however, recurs, and every run of an agent costs model tokens: an agent asked

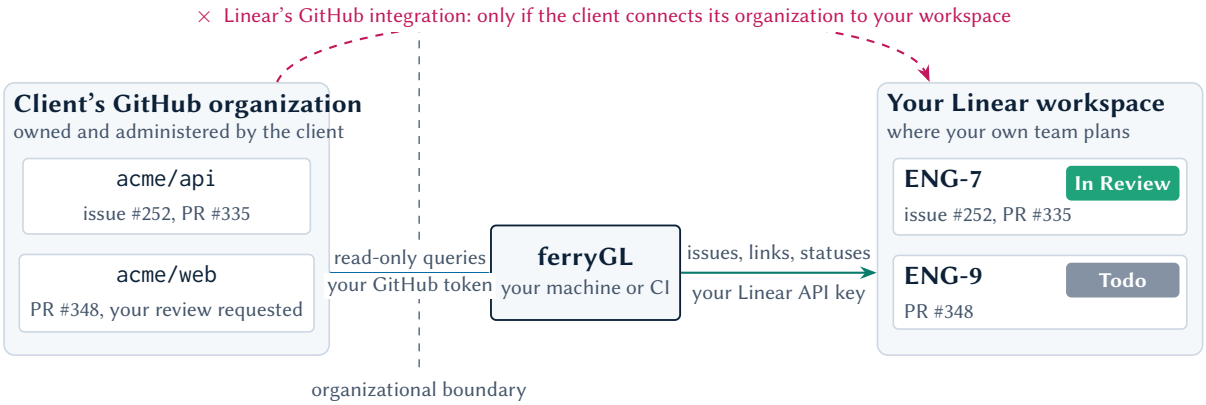


Figure 1. The gap ferryGL fills. Linear's GitHub integration needs the client's organization to connect it, and a GitHub organization can be connected to only one Linear workspace. ferryGL instead reads GitHub with the user's own token, read-only, and writes to the user's own Linear workspace with their own API key. Repository names and issue numbers here and throughout the paper are synthetic.

to keep two systems in step pays to read both again on every run, whether or not anything changed. This paper argues that such work belongs in *token-free automation*: deterministic programs that do a routine, rule-based task without running a language model, and that people and agents can call instead of doing the task themselves. The agent keeps the judgment, such as when to run the task and what to do about an exception, and the program does the repetition, at the cost of API requests rather than model inference. ("Token-free" refers to model tokens throughout; such a program still authenticates with access tokens.) The paper presents one such program, ferryGL, for a task that today falls between tools, and estimates what it saves.

Most software is now built in public or private repositories on hosting platforms such as GitHub, through issues and pull requests [32, 34]. Issue trackers are where teams coordinate that work: they record what needs doing, who is doing it, and how far it has come [9]. Many developers, however, do their work in repositories that their own organization does not own. A consultancy's engineers work in their clients' repositories, a platform team contributes fixes upstream to the open-source projects it depends on, and partners send pull requests to each other. Their own team still plans in its own tracker.

This paper is about one instance of that gap: work on GitHub, tracked in Linear. Linear's GitHub integration links pull requests and issues to Linear issues, but only for repositories whose GitHub organization has connected the integration. That connection must be made on the GitHub side, by an organization owner or, for single repositories, a repository administrator, and "it is not possible to connect multiple Linear workspaces to a single GitHub organization" [45]. A client's organization is often connected to the client's own Linear workspace already, if it connects to one at all (Figure 1).

People bridge the gap in one of three ways today. They copy issues and pull requests by hand, and move each status again whenever something changes on GitHub: repetitive, manual work of the kind site reliability engineering calls toil [61], which adds to the switching between tools and tasks that fragments developers' days [49]. They ask a general-purpose AI agent to do it, since agents that reason and call tools can drive both APIs; but the agent spends tokens on every run to reread the same items and repeat the same moves, and its behavior is not guaranteed to be the same twice [6, 57]. Or they ask the client to connect its organization, which is often not theirs to ask.

ferryGL takes a fourth route. It is a small command-line program that a person runs by hand or on a schedule. Using only that person's GitHub token and Linear API key, it finds the issues and pull requests they are part of in the repositories they list, gives each one a Linear issue assigned to them and linked to GitHub, and keeps the status of that issue in step with GitHub. It never writes to GitHub, never touches a teammate's Linear issue, and never deletes anything. It follows the same rules on every run, shows each change before making it, and spends no model tokens. An AI agent can run it too: instead of doing the job itself, the agent runs ferryGL, reads its plan, and spends its tokens on that alone, 27 to 38 times

fewer on a route of 100 items (Sections 6 and 7.4).

The design rests on one decision: *the destination is the database*. Every Linear issue that ferryGL manages carries a link to its GitHub item, and the link's metadata records the GitHub state that ferryGL last acted on. Linear can find links by URL, so ferryGL needs no state of its own to find its earlier work; and because it compares GitHub with the recorded state rather than with the Linear status, it moves a status only when something happens on GitHub, and leaves the user's own edits alone in between.

Contributions. This paper makes six contributions:

1. The case for token-free automation of routine agent work, and a method to compare an agent doing a task itself, an agent delegating it to a token-free tool, and the tool alone: measure the task's reads and writes, and price them under bracketing assumptions (Sections 6 and 7.4).
2. A statement of the problem of personal, cross-organization, one-way issue synchronization under least privilege, with seven requirements (Section 2).
3. A stateless design that keeps synchronization state in the destination's link metadata, giving idempotent reruns and edge-triggered status updates that preserve manual edits (Section 3).
4. Rules that map a person's roles on GitHub to Linear issues, including pull requests that ride with the issue they fix and the adoption of issues the user made by hand (Section 3).
5. An implementation whose safety rests on eight explicit invariants, a pure planner, and a client interface that cannot express the forbidden writes, with a command-line interface that people and AI agents use alike (Sections 4 to 6).
6. An evaluation with a randomized, model-based simulation that injects write failures, a validated request-cost model, a planner benchmark, and estimates of what an AI agent would spend in tokens, dollars and time doing the job itself and delegating it, on 24 models from 10 providers, released with its scripts and data (Section 7).

2 Background and problem statement

2.1 GitHub and Linear

On GitHub, an *issue* is open or closed; a closed issue has a reason, such as *completed*, *not planned* or *duplicate*. A *pull request* is open, closed or merged, and an open pull request may be a draft. A person can be part of an item in several ways: as its author, as an assignee, as a reviewer who has submitted a review, as a requested reviewer, or as the author of one of its commits. A pull request can declare the issues it closes with keywords such as "Fixes #10" in its description [27], and GitHub search finds issues and pull requests by author, assignee, reviewer, requested reviewer and update date [29]. ferryGL uses GitHub's GraphQL API [35, 36], whose `search` field returns at most 1,000 results [28], and which charges each query rate-limit points: the number of requests needed to fill every connection, divided by 100, with a limit of 5,000 points per hour for a user [26].

In Linear, an issue belongs to a team, has an assignee, and has a status drawn from the team's workflow; each workflow state has a type, such as *unstarted*, *started*, *completed* or *canceled*. An issue can carry *attachments*, which Linear displays as links. An attachment has a URL, a title, a subtitle, and key–value metadata whose "values can be any string or number"; attaching the same URL to the same issue again updates the existing attachment, and the `attachmentsForURL` query returns every attachment to a URL, with its issue [43, 47]. A personal API key acts as its owner [44]. This paper calls attachments *links*, as Linear's interface does.

2.2 Problem statement

A person has a GitHub account g, which can read some repositories owned by others, and a Linear user v in their own workspace. They choose *routes*: each route names some repositories and the Linear team (and, optionally, project) that their work there should go to. The task is to maintain, in each route's team, one Linear issue assigned to v for each GitHub item that g is part of in the route's repositories, linked to the item, with a status that follows the item's state on GitHub, subject to seven requirements:

R1 **Least privilege.** Use only the person's own credentials. Read GitHub, never write to it, and install nothing in the other organization [63].

R2 **Respect for others' work.** Change only Linear issues assigned to v, and never delete anything.

R3 **The person's edits win.** A status set by hand in Linear stays until something happens on GitHub.

R4 **Idempotence without local state.** A rerun never duplicates an issue or a link, and the program keeps no server, database or cache.

R5 **Data minimization.** Copy as little as possible from repositories that belong to someone else [12, 21].

R6 **No surprise writes.** Show every change before making it, and stop before reading or writing anything when the setup is wrong.

R7 **Token-free and deterministic.** Run no language model, so that a run costs API requests rather than model tokens, and the same inputs give the same plan.

Three things are explicitly *not* goals: syncing in both directions, keeping descriptions or comments in sync, and syncing every issue of a repository for a whole team, which is what Linear's own integration does once it is installed. Reacting instantly is not a goal either: webhooks would have to be created in the other organization's repositories, which needs administrator access there [30], so ferryGL polls.

3 Design

3.1 Overview

A run of ferryGL has five steps (Figure 2). It loads and validates the configuration, then resolves every route's Linear setup: the API key's owner, the team, its workflow states, and the project and label if configured. A missing team, state, project or label stops the run before GitHub is read. Then, for each route, it reads GitHub, reads Linear, plans, prints the plan and, only when run as `ferrygl sync --apply`, makes the planned changes in Linear. Without `--apply`, a run performs every step except the writes.

3.2 What is in scope

For each route, ferryGL runs one GitHub search per enabled *role* (Table 1), restricted to the route's repositories and, if configured, to items updated since a given date. It merges the results into one item per URL that keeps every role the person has on it. Each item has a *kind*: `issue`, `pr` (a pull request the person opened) or `involved` (someone else's pull request the person is part of). The kind selects the title template and the status rules. Commenting on an item or being mentioned in it does not bring it into scope, since that would pull in many items touched only in passing.

GitHub search has no qualifier for the author of a pull request's commits [29], so the last role is the one expensive part of a run: ferryGL reads every pull request in scope that the person did not open, with the authors of its first 100 commits, and keeps those with a commit whose author GitHub links to the person's account. Section 7.2 measures what this costs.

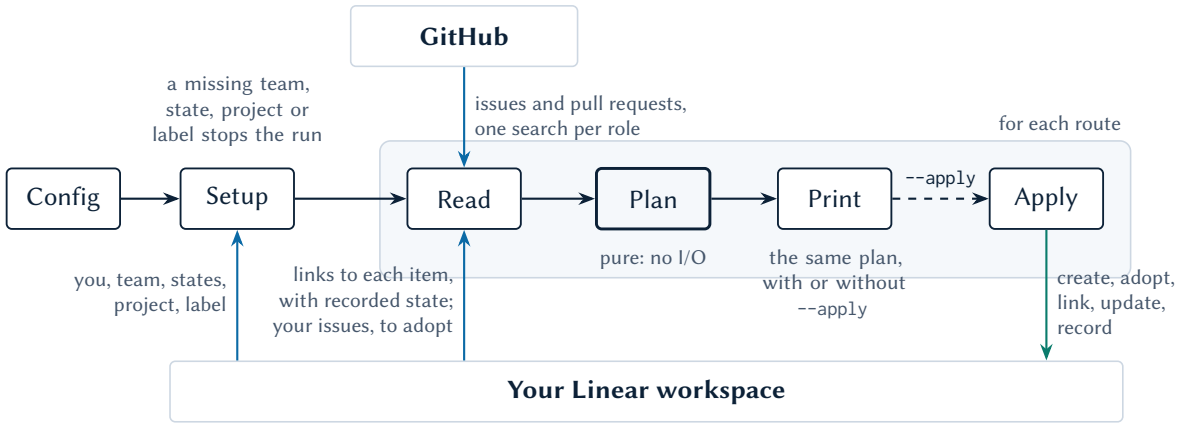


Figure 2. The steps of a run. GitHub is only read. All reads for a route happen before the plan, and all writes after it; the planner itself does no I/O. What a run writes to Linear is what the next run reads back.

Table 1. Roles and the GitHub search behind each, for the login `alice`. Every search is also restricted to the route's repositories (`repo:acme/api`, or `user:acme` for every repository of an owner) and, with `since`, to `updated:>=`*date*.

Flag	Brings in	Search	Kind
`issues_assigned`	Issues assigned to you	`is:issue assignee:alice`	`issue`
`issues_created`	Issues you created	`is:issue author:alice`	`issue`
`prs_created`	Pull requests you opened	`is:pr author:alice`	`pr`
`prs_assigned`	Others' pull requests assigned to you	`is:pr assignee:alice -author:alice`	`involved`
`prs_reviewed`	Others' pull requests you reviewed	`is:pr reviewed-by:alice -author:alice`	`involved`
`review_requests`	Others' pull requests where your review is requested	`is:pr user-review-requested:alice -author:alice`	`involved`
`prs_with_commits`	Others' pull requests with your commits	`is:pr -author:alice`, then a scan	`involved`

3.3 Homes, and state kept in the destination

Each GitHub item has at most one *home*: a Linear issue assigned to v that has a link to the item's URL. ferryGL finds homes by asking Linear for the links to each item's URL (`attachmentsForURL`) and keeping the one on an issue assigned to v. That is the whole of its lookup: it needs no mapping table of its own, and a rerun after a crash, on another machine, or months later finds the same homes. Because the lookup is by link and not by team, a user may move a Linear issue to another team or project and ferryGL still finds it, as long as the new team uses the same status names. Links on issues assigned to anyone else are ignored, so a teammate's issue that links the same pull request is never touched, and reassigning an issue hands it off: if the item is still in scope, the next run creates a new home for the user.

On each home's link, ferryGL records what it needs to remember as flat metadata keys, which fit Linear's string-or-number values (Figure 3). `ferrygl_state` is the GitHub state key ferryGL last acted on, and `ferrygl_synced_at` the start of the run that recorded it; `ferrygl_kind` and the format version `ferrygl_v` complete the record. When ferryGL updates a link it replaces only its own `ferrygl_` keys and keeps any others. These keys, like the state keys below, live in users' workspaces, so they are a compatibility contract: renaming one would make every recorded link look changed.

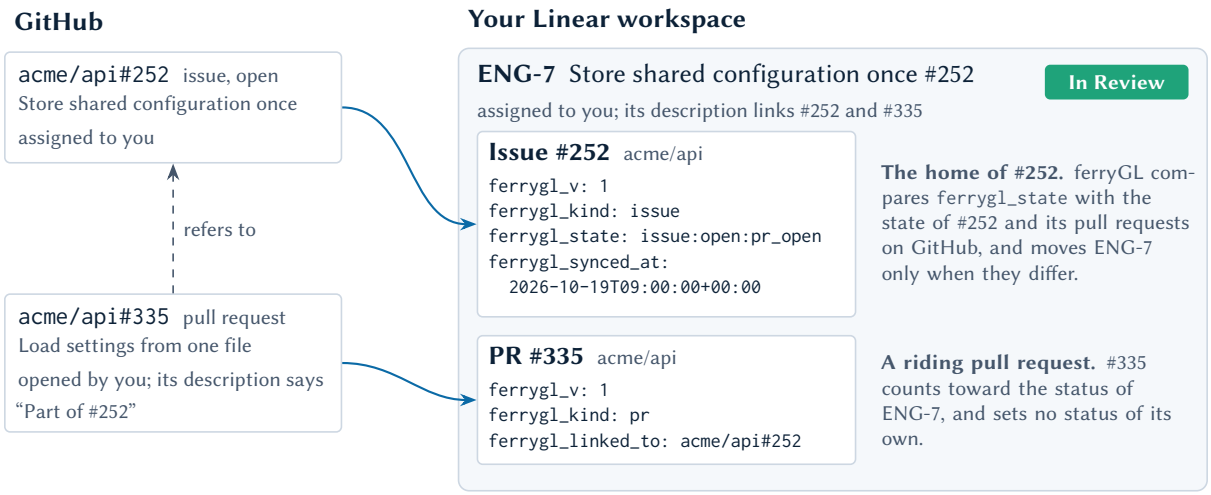


Figure 3. A home and a riding pull request. ENG-7 is the home of issue #252: its link records the GitHub state ferryGL last acted on. Pull request #335 refers to #252, so instead of a Linear issue of its own it gets a link on ENG-7, marked `ferrygl_linked_to`.

3.4 From GitHub state to Linear status

ferryGL summarizes each item's position on GitHub as a *state key* $\sigma(i)$, one of sixteen strings such as `issue:open:pr_open` or `pr:merged` (Figure 4). For a pull request the key follows from its kind and state, and for someone else's pull request also from the person's review roles: a pending review request maps to Todo, even after an earlier review, because it is the person's turn again. An issue's key also depends on the pull requests linked to it (Section 3.6): an open issue whose linked pull request is open and ready for review is In Review, and one whose linked pull requests are all drafts is In Progress.

Each key maps to one of five status categories (Todo, In Progress, In Review, Done, Canceled), whose names in the team's workflow are configurable. The mapping has two forms. $\delta_{\text{new}}(\sigma)$ is the status of a newly created Linear issue, and $\delta_{\text{chg}}(\sigma)$ the status to move to when the state has changed. They agree everywhere except on `issue:open:pr_merged`: the fix has landed but the issue is still open, so a new Linear issue starts in In Review, while an existing one keeps its status, since the work is not done until the issue is closed.

3.5 Edge-triggered updates: the person's edits win

The central rule compares the current state key $s = \sigma(i)$ not with the Linear status, but with the state key ρ recorded on the item's home:

- If $\rho = s$, nothing happens, whatever the Linear status is.
- If $\rho \neq s$ and ρ is recorded, ferryGL moves the status to $\delta_{\text{chg}}(s)$ (an update) and records s. If the issue already has that status, or $\delta_{\text{chg}}(s)$ keeps the status, it only records s (a record).
- If no state is recorded ($\rho = \bot$), this is the link's *first sync*: a link the user added by hand, or one ferryGL just added to an issue the user made. ferryGL then moves the status only to close the issue, to Done or Canceled, when GitHub says the work is done or canceled and the Linear issue is still open. It never reopens an issue, or moves it in any other way, on a first sync.

Algorithm 1 gives the rule in full, with the cases of Sections 3.6 and 3.7.

A status therefore moves only on an *edge* in the GitHub state, relative to a memory that lives in the destination. Figure 5 contrasts this with a level-triggered sync, which would compute the desired status from GitHub on every run and overwrite the person's edits each time. Level-triggered reconciliation is the right model when the source of truth should win continuously, as in the controllers of Borg, Omega

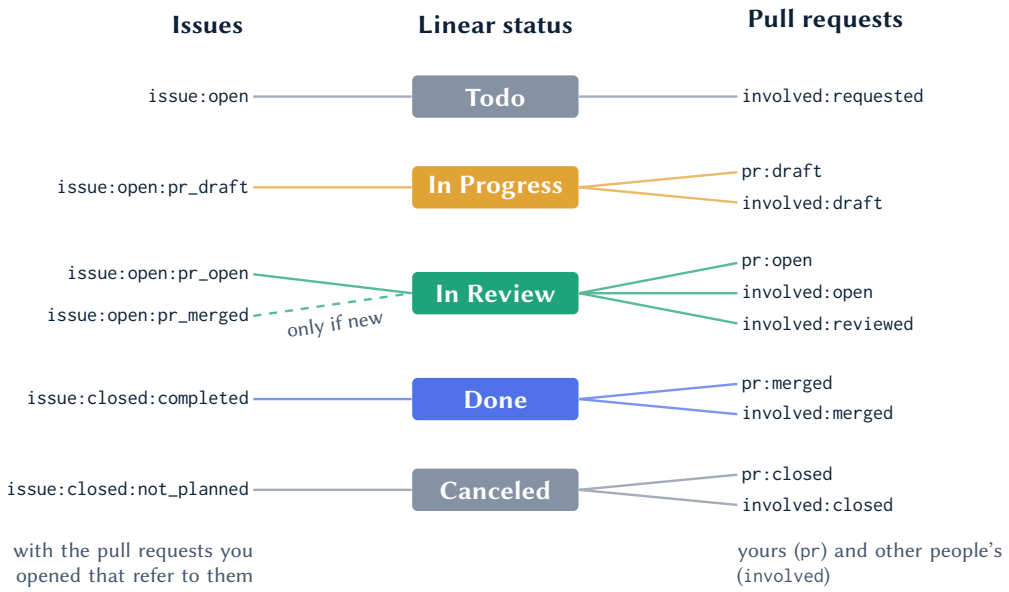


Figure 4. The sixteen state keys and the Linear status each leads to. For issues, the first matching rule wins, from closed to open. `issue:open:pr_merged` sets a status only on a new Linear issue (dashed): when an existing issue's fix merges while the GitHub issue is still open, its status stays as it is.

and Kubernetes [10]. Here two parties own different things: GitHub owns what happened on GitHub, and the person owns their own tracker. Recording the last synchronized state and comparing against it is how file synchronizers tell which replica changed [7, 59]. ferryGL applies the same update detection to the source only, and treats the Linear status as the person's between edges.

Two properties follow, and Section 7 tests both.

Edits persist. If a run completes its writes, every home it touched records the state key it acted on. If, by the next run, the key ferryGL computes for an item is unchanged, the first branch applies and the home's status is left alone, whatever the user did to it in between. (The key of an issue depends on its linked pull requests, so a new review request or a merge is an edge; Section 7.1 shows one case where the key changes without anything new on GitHub.)

Quiescence. If neither GitHub nor Linear changes after a run that completed its writes, the next run plans no writes: every item in scope has a home that records its current key, and every rider is linked. Reruns are therefore safe to schedule as often as wanted.

3.6 Pull requests that ride with an issue

An issue and the pull request that fixes it are one piece of work. When a pull request the person opened refers to one of their synced issues in the same route, it does not get a Linear issue of its own: it *rides* with the issue, and ferryGL links it to the issue's home instead, with `ferrygl_linked_to` naming the issue (Figure 3). A pull request refers to an issue when GitHub lists the issue as one it will close, or when its title or description mentions it as `#252`, `acme/api#252` or by URL. The issue's state key then accounts for its linked pull requests, so the home follows the issue and its fix together: In Progress while the fix is a draft, In Review when it is open, and Done only when the issue itself is closed. The rider's link sets no status of its own, so each Linear issue has one status source.

A few cases need care. A pull request that already had a home of its own before it referred to the issue keeps it, and still counts toward the issue's status. Conversely, an issue that has no home while its pull request does *joins* the pull request's home, as long as the pull request refers to no other synced issue. A rider's link stays when the reference is later edited out, and it keeps counting toward the issue's

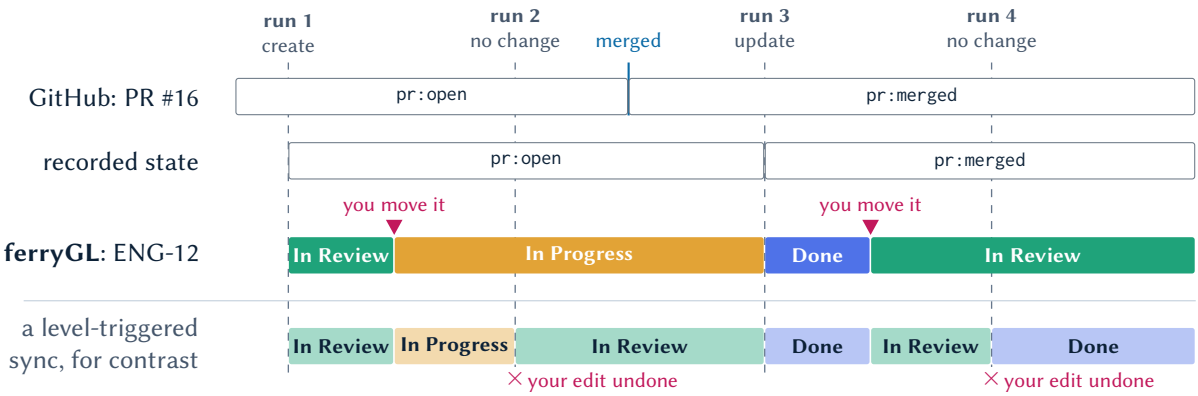


Figure 5. Edge-triggered updates for one pull request and its Linear issue. Run 1 creates ENG-12 in In Review and records `pr:open`. The user moves it to In Progress; run 2 sees the recorded state still matches GitHub and changes nothing. Run 3 sees the merge and moves it to Done. The user then reopens it, and run 4 leaves it alone. A level-triggered sync (bottom) would undo both edits.

status. When the issue leaves the person's scope (they are unassigned, say) while the pull request is still in it, the issue's home stays as it is: the rider does not move it on its own. Someone else's pull request always gets its own home, even if it refers to the person's issue. Finally, a pull request that refers to several of the person's issues rides with each of them, so its URL is linked from each of their Linear issues, as a rider; it is still the home of none.

3.7 Adopting issues the person made by hand

People often create a Linear issue for a GitHub item themselves before they install ferryGL. Before creating a new issue, ferryGL therefore looks for one of the person's issues in the route's team that already mentions the item: its URL in the description or in a link, or, for a route with a single repository, a title ending in *#number* with no link to any other GitHub item. A candidate already linked to a different GitHub item, or already home to another item, never matches. Exactly one match is *adopted*: ferryGL adds its link and applies the first-sync rule, so the adopted issue's status moves only to close it. Several matches are *skipped*, with their identifiers: the person adds the link to the right one by hand, and the next run picks it up.

3.8 Writes, failures and retries

Each change is applied in a fixed order: the status first, then ferryGL's comment (if comments are on), then the links. The recorded state therefore changes last: if the status update or the comment fails, the old state stays recorded, and the next run plans the same change again. Each change is applied independently, so one failure does not stop the others; the run then exits with status 1.

Retries follow the idempotency of each request (Table 2) [22, 37]. Rate-limited responses (HTTP 429 [55], or GitHub's 403 with `Retry-After`) are always retried, after the requested wait. Network errors, timeouts and HTTP 502, 503 and 504 are retried only for idempotent requests: every read, a status update, and a link create or update, which Linear makes idempotent per issue and URL [43]. Creating an issue or a comment is not idempotent: if the connection fails, Linear may have created it anyway. ferryGL reports the error instead of retrying. If the issue was created, the next run finds it through the GitHub URL in its description and adopts it, so no duplicate is created; a comment, however, may be posted twice. Linear's create inputs accept an identifier chosen by the client [47], which could make these retries safe as well (Section 9).

Algorithm 1 Planning one item i, with its linked pull requests L and riders R (Section 3.6). $A(u)$ are the links to URL u returned by Linear, C the user's issues in the route's team, and v the user.

$s \leftarrow \sigma(i, L)$
$h \leftarrow$ the first link in $A(url(i))$ on an issue assigned to v, if any ▷ the home
if h exists **then**
    $\rho \leftarrow$ the state recorded on h, or $\perp$
    if $\rho \neq s$ **then**
        if $\rho = \perp$ **then** ▷ first sync: only close
            $c \leftarrow \delta_{\text{chg}}(s)$ if it is Done or Canceled and h's issue is open, else none
        else
            $c \leftarrow \delta_{\text{chg}}(s)$
        emit update(h, c) if c differs from the issue's status, else record(h); both record s on h
    emit link(r) for each $r \in R$ not yet linked to h's issue
else
    $M \leftarrow \{\, m \in C$ unclaimed, mentioning i or a rider, and linked to no other GitHub item $\}$
    if $|M| > 1$ **then**
        emit skip(i, M) ▷ the user decides
    else if $|M| = 1$ **then**
        emit adopt(i, m) under the first-sync rule, and claim m
    else
        emit create(i) with status $\delta_{\text{new}}(s)$, linking i and R

Table 2. Retry policy. Each request gets up to three attempts. Waits honor `Retry-After`, up to 60 seconds, and are otherwise 2 and then 4 seconds. A request times out after 30 seconds, which counts as a network error.

Request	Rate limited (429, or 403 with `Retry-After`)	Network error, timeout, or 502, 503, 504
GitHub and Linear reads	retried	retried
`issueUpdate`, `attachmentCreate`, `attachmentUpdate`	retried	retried
`issueCreate`, `commentCreate`	retried	reported; adopted or retried next run

4 Implementation

ferryGL is a Python package of about 1,900 lines in eight main modules (Table 3), with one runtime dependency, the HTTP client HTTPX [20]. It supports Python 3.11 and later. Its structure follows the "functional core, imperative shell" pattern [8, 53]: `plan.py` is a pure function from the GitHub items, the Linear links and candidates, and the configuration to a list of changes, and everything that talks to the network lives around it. All types passed between modules are frozen dataclasses.

Configuration. The configuration is one TOML file [60] (Section A). Only the routes are required. Unknown tables and keys are errors, so a misspelled setting cannot be silently ignored, and every error names the file and the setting. A JSON Schema [71, 72] published on the project's website gives editors completion and inline checks; a test keeps it in step with the parser.

Commands and output. `ferrygl check` confirms that the Linear key works, that the GitHub token can see every repository, and that each route's team, states, project and label exist; it writes nothing. `ferrygl sync` prints the plan, one line per change, and `ferrygl sync --apply` prints the same plan and makes it:

```
acme/api#252  issue     update  ENG-7 Todo -> In Review
acme/api#335  pr        link    -> ENG-7
```

Table 3. Modules of ferryGL 0.2.1, with their length in lines.

Module	Responsibility	Lines
`plan.py`	Pure planning: GitHub items and Linear state to changes; state keys	464
`sync.py`	`check` and `sync`: resolve routes, plan, print, apply	318
`config.py`	Parse and validate `ferrygl.toml` into frozen dataclasses	288
`linear.py`	Linear GraphQL reads, and the five mutations ferryGL uses	279
`github.py`	Tokens, one search per role, the commit scan, references	272
`model.py`	Shared types: items, Linear issues, links, changes	124
`cli.py`	Arguments, the Linear API key, exit statuses	92
`_http.py`	JSON over HTTP with retries, for both clients	81

```
acme/api#360  issue     create  "Cache the session lookup #360" -> Todo
acme/api#348  involved  adopt   -> ENG-9
4 changes (1 create, 1 adopt, 1 link, 1 update). Run with --apply to make them.
```

The exit status is 0 when everything worked, 1 when a route could not be read or a write failed, 2 for a setup or configuration error (in which case nothing was read from GitHub or written to Linear), and 130 when interrupted. With comments turned on, every change also posts a Linear comment that starts with the marker `(BY-FGL)`, so people can tell ferryGL's changes from their own and filter on them.

Requests. Per route, a run makes one GitHub query for the token's login, then one paged search per enabled role, at 50 results per page, or 25 pull requests per page for the commit scan. On the Linear side, it looks up the user, each route's team, states, project and label, then the links to up to 25 URLs per request, and, only when some item has no home yet, the user's issues in the team, 50 per page. With `--apply`, each write is one request. Linear allows 2,500 requests an hour for an API key [46]; a first run that creates a few hundred issues uses a noticeable share of that, and later runs write only what changed.

Tests. The repository has 155 tests that run in under a second. Each planning test states a scenario in plain terms (what is on GitHub, what is in Linear, the expected changes); the clients are tested against `httpx.MockTransport`; and `check` and `sync` run end to end against fakes. All test data is synthetic.

5 Safety and privacy

ferryGL usually runs in repositories that belong to someone else, often private ones whose contents are covered by an agreement, so its safety rules matter more than its features. The repository's contributor guide states them as eight invariants, which human contributors and coding agents alike must not break without the project owner's decision. Table 4 lists each one with the mechanism that enforces it and the evidence from Section 7.

Safety by interface. Two invariants hold by construction rather than by care. The GitHub client has no code path that sends a mutation. The Linear client exposes exactly five mutations: `issueCreate`, `issueUpdate` with only a state, `attachmentCreate`, `attachmentUpdate` and `commentCreate`. No method can delete an issue, a link or a comment, or edit an issue's title or description or a comment, so no bug in the planner can make ferryGL do so.

Credentials. ferryGL sends the GitHub token only to `api.github.com` and the Linear key only to `api.linear.app`, and stores neither. A read-only GitHub token is enough, and preferred; some organizations must approve fine-grained tokens before they can read their repositories, which `ferrygl check` reports as a repository it cannot see. The Linear key acts as the user, so ferryGL's own rules, not

Table 4. The eight invariants, how the design enforces them, and the evaluation evidence.

Invariant	Mechanism	Evidence
I1. GitHub is read-only	The GitHub client only sends GraphQL queries	No mutation among all requests (Section 7.2)
I2. Only the user's Linear issues; nothing deleted or rewritten	Writes target homes or new issues assigned to the user; the Linear client has no delete, and its update sets only the status	*ownership* (Table 5)
I3. The user's edits win	Edge-triggered updates against the recorded state; first syncs only close	*status source*, *edits win*
I4. One home per item, one status source per Linear issue	Lookup by link; claimed candidates; riders set no status	*one home*, *coverage*
I5. Idempotent reruns	Homes found by link; only idempotent requests retried	*quiescent*, *recovery*
I6. A pure planner	All reads before planning, all writes after	*order independent* (one finding)
I7. Copy as little as possible	Only titles, numbers, repositories, URLs, states and roles by default	*minimal data*
I8. No surprise writes	Preview by default; setup checked before reading GitHub	*dry run*

the key's scope, keep it to the user's issues; a key with limited permissions must still allow the writes ferryGL makes.

What reaches Linear. By default, ferryGL writes the GitHub title and number (the title template is configurable, and can leave the title out), the repository, the item's URL, its state key, and the user's role as implied by the kind. It reads descriptions to find the issues a pull request refers to, but copies them only if the user turns that on, once, when the issue is created. It never copies GitHub comments, reviews, labels, code, or other people's names or logins, including the authors of pull requests and commits. Its own comments name only GitHub items, states and Linear status names. It keeps nothing on the user's machine: no database, cache or log file of its own.

6 Token-free automation for AI agents

Many people who would use ferryGL already work with an AI agent. An agent asked to keep Linear in step with GitHub can do that job in two ways. It can do the job itself: search GitHub for each role, look up each item in Linear, apply the rules of Section 3, and make the writes, reading every item on every run. Or it can delegate the job to a token-free tool: run `ferrygl sync`, read the plan, run `ferrygl sync --apply`, and report. Its tokens then go to what ferryGL prints, a line per change and `No changes.` on a quiet run, instead of to every item in scope. Section 7.4 measures the difference: on a route of 100 items, 27 to 38 times fewer tokens, and on 800 items 122 to 934 times fewer.

An interface agents can use. ferryGL needs nothing agent-specific: any agent that can run a command can use it, through the same interface a person uses. Three of its properties make that safe and cheap. A run previews by default: `ferrygl sync` writes nothing, so an agent can show the plan to its user, or check it, before it runs `ferrygl sync --apply`. The output is compact and regular: one line per change, with the item, its kind, the action and its details, then a one-line summary, and every `skip` line says what decision it needs. And the exit status says what happened: 0 when everything worked, 1 when a route or a write failed, 2 when the setup is wrong and nothing was read or written, and 130 when interrupted. ferryGL's documentation has a page for agents, with instructions a user can give their agent, and its website publishes an `llms.txt` index of the documentation for language models [39]. Instructions as short as these are enough:

```
To sync my GitHub work into Linear, run `ferrygl sync` and show me the plan.
Run `ferrygl sync --apply` only after I approve it. If a line says skip,
tell me what ferryGL needs. If the exit status is 2, report the error and stop.
Never change GitHub or Linear yourself to do this.
```

Least privilege for agents. Delegating also narrows what an agent can do. An agent whose only access to GitHub and Linear is ferryGL cannot write to GitHub, change a teammate's Linear issue, or delete anything, whatever it is asked or gets wrong: the invariants of Table 4 hold for every caller, and ferryGL's preview shows every change first [63]. The agent needs no GitHub or Linear tools of its own.

A general pattern. The same split applies wherever an agent is asked to do routine, rule-based work again and again: keeping trackers in step, triage by fixed rules, recurring reports. Once a task's rules are known, writing them into a program once moves the task's marginal cost from inference to API requests, and makes its behavior repeatable. This complements tool-using agents [64, 70] and agents that write tools for themselves [11]: the tool here is written once, tested, and shared, and the agent's role is to decide when to call it and to read what it reports. It is an instance of the compound AI systems that combine models with conventional components [77]; it complements other ways of reducing inference cost, such as routing requests to cheaper models [13]; and it answers calls to weigh agents' cost alongside their accuracy [42, 74] and to count efficiency as a goal [65].

7 Evaluation

The evaluation asks four questions. Do the safety invariants hold over long, random histories in which GitHub changes, the user edits Linear, teammates link the same items, and writes fail (Section 7.1)? How many requests and rate-limit points does a run cost, and what dominates (Section 7.2)? Does the planner scale (Section 7.3)? And what would an AI agent spend to do the same job, by itself or by delegating it to ferryGL, and how does that vary across models and providers (Sections 7.4 and 7.5)? The scripts and their output are in the repository, under `docs/paper/` (Section B).

7.1 Randomized, model-based simulation

Method. The simulation runs ferryGL's own reading, planning and writing code, unchanged, against two in-memory stand-ins with the same methods as its GitHub and Linear clients. The GitHub stand-in answers each role's search from a model of issues and pull requests, in the shape of GitHub's GraphQL nodes, so ferryGL's result parsing and reference detection run too. The Linear stand-in keeps issues, links and comments, and follows Linear's documented behavior: links are returned by URL across the workspace, up to 20 per URL, and attaching a URL an issue already has updates that link. As in property-based testing [16], each scenario is generated at random from a seed: one or two repositories, a random configuration (each role on with probability 0.85, comments on or off), and a few initial items. Then, for 12 steps, random events happen and ferryGL runs. GitHub events open and close issues, open, ready, draft, merge, close and reopen pull requests, request, give and withdraw reviews, assign and unassign the user, add and remove references, and push commits. Linear events have the user move statuses, hand issues to a teammate, link items by hand, and write issues that mention items, and have a teammate link the same items and the user add their own metadata to ferryGL's links. Three in ten steps change only Linear. Each step runs ferryGL four ways: a preview, an apply in which 5 % of writes fail (half of them *lost*, reaching Linear but losing their response, and half *rejected*), a clean recovery run if any write failed, and a final clean rerun.

After every run, the simulation checks the properties of Table 5 against the stand-ins' state and the writes ferryGL sent. GitHub descriptions in the model carry a secret marker and mention other users, so *minimal data* can check that no write carries either, and that no comment carries a GitHub

Table 5. Properties checked by the simulation, how many times each was checked, and how many checks failed. A check is one evaluation of the property on one run's state, writes or plan.

Property	Statement	Checks	Failed
dry run	A run without `--apply` writes nothing	24,000	0
quiescent	A rerun with nothing new on GitHub or in Linear writes nothing	24,000	0
one home	Each GitHub item is linked as a home from at most one of the user's issues	80,923	0
ownership	Every write targets an issue assigned to the user	49,178	0
status source	A status moves only with a new recorded state, or to close on a first sync	4,742	0
edits win	After a step that changed only Linear, every status the user set is still set, unless a first sync closed the issue	4,098	0
coverage	Every item in scope is linked from one of the user's issues, unless it, or the issue it rides with, was skipped	86,168	0
consistent	A pull request's home records its current state and, unless the user moved it, has the status Figure 4 gives	75,204	0
foreign metadata	Metadata keys that are not ferryGL's survive every update	11,302	0
minimal data	No write carries a description or another user's login, and no comment a title	49,178	0
recovery	After failed writes, one clean run succeeds and converges	4,278	0
order independent	The plan does not depend on the order of Linear's lookup results	14,065	29

title. *Order independent* is a metamorphic relation [14, 15]: planning the same inputs with every list of Linear lookup results reversed must give the same changes, since Linear promises no particular order.

Results. 2,000 scenarios produced 55,538 events, 74,139 runs and 49,178 writes, of which 2,250 failed. The plans included 11,421 creates, 5,734 updates, 1,598 links, 821 records, 412 adoptions and 133 skips. The whole simulation takes about ten seconds. Table 5 shows the results: every safety property held in every check. Write failures left 776 Linear issues created without their links; the next clean run adopted all 776 of them through the URL in their description, and every recovery converged in one run.

A finding: order dependence. The metamorphic check failed in 29 of 14,065 non-trivial checks, all with the same cause. A pull request that refers to two of the user's issues rides with both, so its URL is linked from both of their Linear issues. If one reference is later edited out, the planner looks up the pull request's links and takes the first one on the user's issue as the pull request's own. When that first link is on the issue it still refers to, the pull request rides with that issue only, and stops counting toward the other issue's status: that issue's state key changes, say from `issue:open:pr_open` to `issue:open`, and its Linear issue moves back to Todo. When the first link is on the other issue, the sticky-link rule of Section 3.6 applies, as documented, and nothing moves. Since Linear does not promise an order for these results, the outcome is not determined. The case is narrow (a pull request that refers to several of the user's issues and then drops one reference) and moves only the status of the user's own issue, but it breaks the documented rule and the purity of the planner's inputs. A fix is to choose among a pull request's links deterministically, preferring a link on the home of an issue it still refers to, and to keep counting the pull request toward the status of every issue whose home has its link.

7.2 Request cost

Method. The cost experiment runs ferryGL's real GitHub client against a mock of GitHub's GraphQL endpoint, built on `httpx.MockTransport`, that pages results as GitHub does and stops at 1,000 results per search. The repository holds a fixed set of the user's own items (12 issues, 8 pull requests) and a growing number of other people's pull requests; one in ten requests the user's review, one in ten was

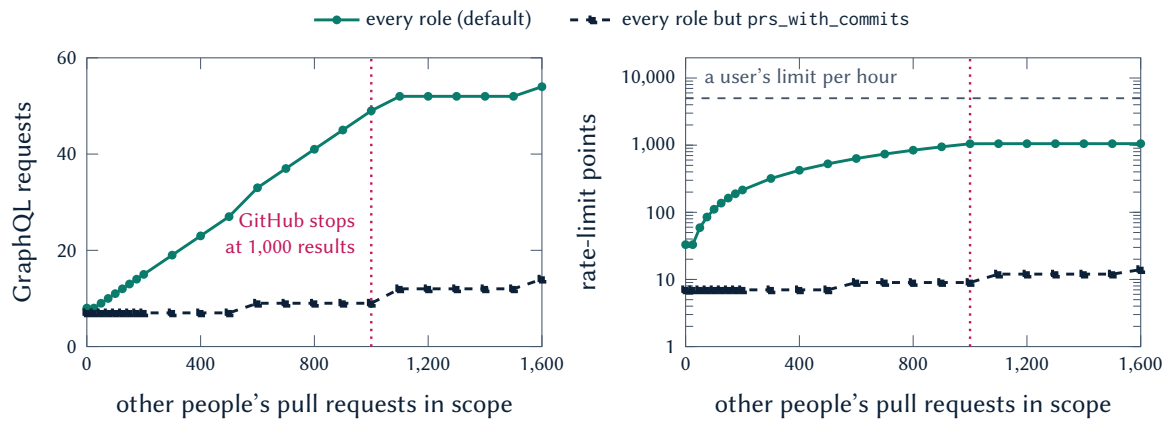


Figure 6. GitHub requests (left) and rate-limit points (right, log scale) for one route, as the number of other people's pull requests in scope grows. The commit scan (`prs_with_commits`) dominates the cost. GitHub returns at most 1,000 results per search, so past that point the cost flattens and the scan misses pull requests.

reviewed by them, one in twenty is assigned to them, and one in twenty has one of their commits. For each size, the experiment counts the requests ferryGL sends and the rate-limit points GitHub would charge, using GitHub's documented cost formula [26]. A search page of 50 results with their closing references costs 1 point. A commit-scan page of 25 pull requests, each with 100 commits of up to 5 authors, needs $1 + 25 + 25 + 25 \times 100 = 2{,}551$ requests to fill, and costs 26 points.

Results. The request count ferryGL sent matched a closed-form model at every size, and none of the requests was a mutation. Figure 6 shows the result. Without the commit scan, a route with 1,000 other pull requests in scope costs 9 requests and 9 points. With it, the same route costs 49 requests and 1,049 points, about a fifth of a user's hourly 5,000; the scan accounts for 40 of the requests and 1,040 of the points. Beyond 1,000 pull requests, GitHub's limit on search results truncates the scan: the cost stops growing, but pull requests past the 1,000th are no longer checked, so some of the user's commits go unseen. This is why ferryGL's documentation recommends setting `since`, which bounds every search by update date, and turning `prs_with_commits` off when the role is not needed.

7.3 Planner performance

The benchmark times the planner alone, on synthetic routes of 10 to 10,000 items, four in ten of them issues, three in ten the user's pull requests (most referring to an issue) and three in ten other people's. In one series every item has a home, one in ten with an out-of-date recorded state; in the other, a fifth of the items are new, so the planner also matches them against the user's issues for adoption. Figure 7 reports the median of seven runs on an AMD Ryzen 9 9950X3D with CPython 3.12. Planning grows linearly when every item has a home, and takes about a millisecond for a few hundred items. Adoption matching compares each new item with each candidate, so it grows faster than linearly, but stays at about a third of a second for 10,000 items, which is more than any person is part of. A run's time is dominated by its network requests.

7.4 What an AI agent would spend, and what delegating to ferryGL saves

Method. An AI agent with access to GitHub and Linear could do ferryGL's job itself, or delegate it to ferryGL (Section 6). No agent was run for this paper. Instead, the experiment measures the work each strategy implies, and prices it. It runs ferryGL over 48,000 runs of random histories generated as in Section 7.1, without injected failures, and over single routes of 10 to 800 items, through stand-ins that record every call. For an agent doing the job itself, they record the size of each result it would have to read and of the arguments of each call it would have to send, if its tools returned exactly what ferryGL's

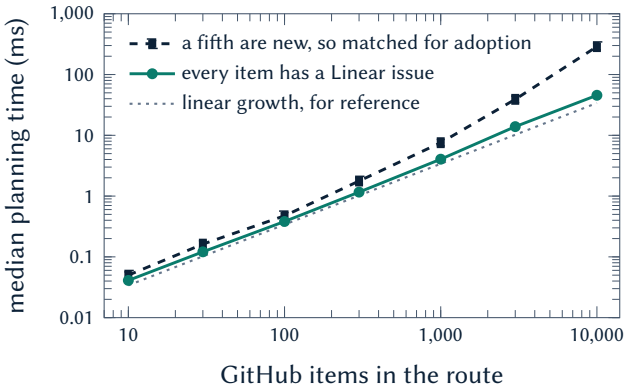


Figure 7. Median planning time for one route. Every item having a home is the steady state. When a fifth of the items are new, matching them against the user's issues for adoption grows faster than linearly.

queries return. For a delegating agent, they record what ferryGL prints, which is all it reads. A cost model then prices each agent loop:

- An agent doing the job itself starts each run with ferryGL's rules as instructions (its sync rules and link-metadata convention, about 9,700 characters), nine tool definitions shaped after ferryGL's reads and writes (about 2,600 characters), the 286-token tool-use system prompt [4], and a short request to sync: about 3,361 tokens. A run has three dependent steps (read GitHub and the Linear setup; look up the links to each item; write) and a closing turn, and a step takes one turn per batch of parallel tool calls. Letting all of a step's writes share a turn favors the agent, since a link can only be added once its issue exists.
- A delegating agent starts with a five-sentence instruction, which lets it apply the plan without asking, and the bash tool, whose definition adds 325 tokens to the tool-use system prompt [4]: about 735 tokens. It runs `ferrygl sync`, reads the plan, runs `ferrygl sync --apply`, reads the result, and reports, in three turns.
- Prompt caching is on: each turn reads the previous prompt from the cache and writes only what is new. Scheduled runs are hours apart, so each run starts with a cold cache.
- Prices are Anthropic's published rates for Claude Opus 5.5, Sonnet 5.5 and Haiku 5.5, including Haiku 5.5's higher rates for prompts over 100,000 tokens [4].

Two sets of assumptions bracket how an agent behaves (Table 6), and both strategies are priced under both. Neither set describes a measured agent. The *lean* assumptions give the agent every advantage, so they approach the least an agent could spend. The *plausible* ones are a middle estimate, not a worst case: general-purpose GitHub and Linear tools return fuller records than ferryGL reads, so an agent using them would read more. In the route experiment the commit scan is off, and each run finds a twentieth of the items changed.

Results. On a route of 100 items, an agent doing the job itself would read and write 91,620 to 217,857 tokens per run; an agent delegating it to ferryGL, 3,364 to 5,769; and ferryGL on its own, none (Table 7, Figure 8). On Claude Opus 5.5, doing the job itself costs $10.91 to $18.48 a month at two runs a workday, or $43.64 to $73.91 a month with hourly runs; delegating it costs $0.71 to $1.95, or $2.85 to $7.79 a month with hourly runs. Four findings stand out.

- **Delegating cuts the tokens many times over, more the larger the route.** At 10 items, a delegating agent uses 7 to 8 times fewer tokens than one doing the job itself; at 100 items, 27 to 38 times fewer, at 9 to 15 times less cost on Opus 5.5; at 800 items, 122 to 934 times fewer tokens, and 77 to 79 times less cost. After the first run, a delegating agent's cost grows only slowly with the route, because ferryGL prints a line per change, not per item: 3,364 to 5,769 tokens at 100 items,

Table 6. The two sets of assumptions about how an AI agent works. Neither describes a measured agent; together they bracket one. Characters per token are measured (Table 8): over its seven distinct tokenizers, the most characters per token for lean and the median for plausible, for each kind of text.

Assumption	Lean	Plausible
Average length of a GitHub description it reads	0 characters	500 characters
Tool calls per turn	all of a step's	at most 5
Reasoning tokens per turn, billed as output	100	500
Characters per token: JSON / prose, for models without a public tokenizer	2.93 / 4.06	2.81 / 4.02
Shared by both	tools that return only the fields ferryGL reads; ferryGL's rules as instructions; prompt caching, cold at the start of each run; each provider's published prices; a model's own tokenizer where it is public (Table 8)	

Table 7. What an AI agent would spend on a route of 100 items, for a run that finds 4 changes to make, doing ferryGL's job itself or delegating it to ferryGL. Monthly costs assume two runs a workday. Tokens per run do not depend on the model. Estimates from the cost model of Section 7.4; ferryGL alone measured.

		Doing the job itself		Delegating to ferryGL	
Model	Assumptions	Tokens per run	Per month	Tokens per run	Per month
Claude Opus 5.5	lean	91,620	$10.91	3,364	$0.71
	plausible	217,857	$18.48	5,769	$1.95
Claude Sonnet 5.5	lean	91,620	$5.46	3,364	$0.36
	plausible	217,857	$9.24	5,769	$0.97
Claude Haiku 5.5	lean	91,620	$0.28	3,364	$0.018
	plausible	217,857	$0.50	5,769	$0.049
ferryGL alone		0 tokens, $0			

and 4,991 to 7,414 at 800. What remains is mostly its own instructions and reasoning.

- **A run that finds nothing new costs almost as much.** An agent doing the job itself can learn that nothing changed only by reading everything again. A run of the same route that finds nothing new reads fewer tokens (54,966 to 106,048) but still costs $0.22 to $0.36 on Opus 5.5, close to the cost of a run that finds 4 changes to make. In the random histories, where every other run finds nothing new, those runs account for 46 % of the agent's spend; on a schedule, most runs find nothing new.
- **An agent's own cost grows faster than scope.** Its context grows with every item read, and is read again on every turn. At 400 items, the plausible agent reads and writes 1,941,133 tokens per run, $73.01 a month on Opus 5.5 at two runs a workday. Haiku 5.5's rates rise fivefold once a prompt passes 100,000 tokens, which shows as a jump between 100 and 800 items, depending on the assumptions. The first run, which creates every Linear issue, costs $0.69 to $1.70 on Opus 5.5 for 100 items.
- **A smaller model costs less, at a risk not measured here.** Sonnet 5.5 halves the cost of Opus 5.5, and Haiku 5.5 cuts it by more than an order of magnitude at modest scope. Whether a smaller model applies ferryGL's rules reliably is exactly what this estimate cannot say; a delegating agent needs only to run two commands and read their output.

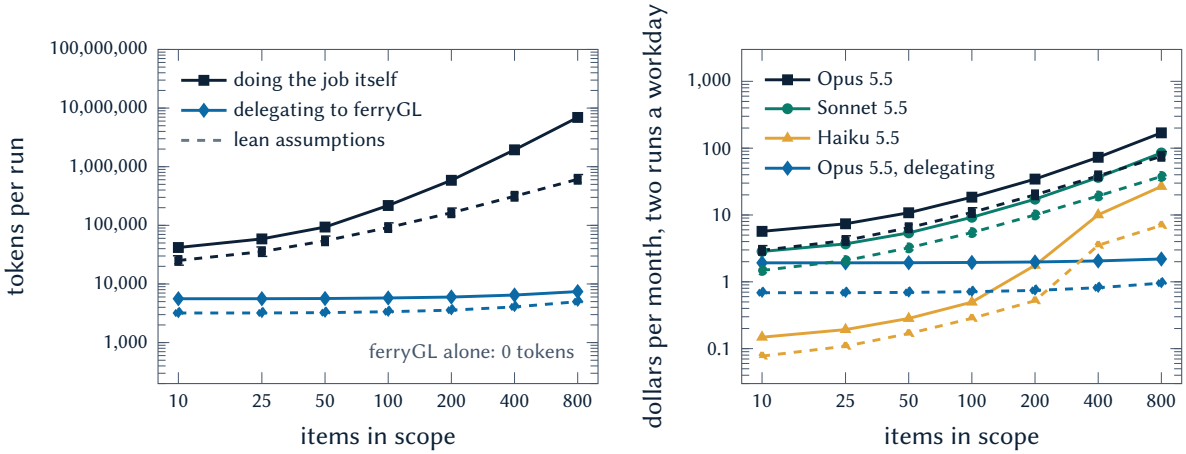


Figure 8. What an AI agent would spend, by the size of the route, doing ferryGL's job itself or delegating it to ferryGL: tokens per run (left) and dollars per month at two runs a workday (right), for runs that find a twentieth of the items changed. Solid lines use the plausible assumptions, dashed lines the lean ones. Both axes are logarithmic. ferryGL alone spends no tokens.

Interpretation. For one person with a modest route, an agent's bill is small in absolute terms: $5.46 to $18.48 a month on Sonnet or Opus at two runs a workday, and under a dollar on Haiku. The case for token-free automation rests less on the size of the bill than on its shape. An agent's own cost recurs on every run, including the runs that find nothing to do; it grows with the route and the schedule; and it multiplies with every person who runs it. It also buys no guarantees: the estimate assumes the agent applies the rules correctly every time, which language models do not promise [6, 57], and a run that gets one wrong (a status moved over the user's edit, a duplicate issue, a teammate's issue changed) costs more than its tokens. Delegating shows that the two are not alternatives: an agent that runs ferryGL keeps its judgment for the decisions that need it, such as when to run and what to do about a skipped item, spends a few thousand tokens per run on them after the first, growing only slowly with the route, and inherits ferryGL's guarantees.

Limits of the estimate. This is a model of agents, not a measurement of them. Reasoning tokens per turn, calls per turn and turns per run are assumptions, chosen in the lean case to favor the agent; every provider is given Anthropic's tool-use overhead (286 tokens, and 325 more for the bash tool), since not all publish theirs. Token counts use the characters per token that public tokenizers give on the workload's own text (Table 8); Claude's tokenizer is not public, so the measured maximum and median stand in for it. The agents' correctness, determinism and latency are not measured. Running real agents with these tools over the same histories would replace the assumptions with measurements, and would also count how often an agent doing the job itself breaks the invariants of Table 4.

7.5 Across 24 models from 10 providers

Method. Section 7.4 priced agents on Anthropic's models. This experiment repeats it on 24 current models from 10 providers: for each provider, its flagship and at least one cheaper model, among them the leading open-weight models as served by their makers (Table 9). Everything but the model is held fixed: the same measured workload (one user's route of 100 items, on a run that finds 4 changes to make, with 21 requests to GitHub and Linear), the same three strategies, and the same lean and plausible assumptions (Table 6). Three inputs vary by model, each from a published source read on 19 October 2026; the schedule and the size of the team vary for every model.

- **Prices.** Each provider's standard API prices for input, cache writes, cache hits and output, under its own caching rules: whether a cache write costs more than input, the shortest prompt it caches, and any higher price for long prompts [1, 3, 4, 18, 31, 50–52, 56, 73, 78]. Where a price is set to

Table 8. Characters per token of the text an agent reads and writes on a route of 100 items, measured with each public tokenizer of a model family in the comparison. OpenAI's `o200k_base`, used by its GPT-5 models, is shown for reference; GPT-6's tokenizer is not published.

Tokenizer	JSON tool data	Instructions	ferryGL's output
Mistral Small 4	2.38	3.93	3.48
Qwen3.8 Max, Qwen3.8 Flash	2.50	3.89	3.48
DeepSeek V4 Pro, DeepSeek V4.1 Flash	2.75	4.00	3.68
GLM-5.3, GLM-5.3-Flash	2.81	4.02	3.70
OpenAI o200k_base	2.86	4.02	3.66
MiniMax-M3	2.92	4.05	3.66
Kimi K3, Kimi K2.7 Code	2.93	4.06	3.66

rise on an announced date, the later price is used, since the comparison looks a year ahead; where some hours cost less, the full price is used.

- **Tokenization.** The open-weight models' own tokenizers, run on the text of the workload (Table 8). JSON tool results, most of what an agent reads, come to only 2.4 to 2.9 characters per token, well below the four characters per token often assumed for prose. Claude, GPT-6, Gemini, Grok, Mistral Large 4 and MiniMax-M3.1-Flash have no public tokenizer, and take the measured median (plausible) or maximum (lean), per kind of text.
- **Speed.** Each model's output speed and time to first answer token on its maker's API, as measured by Artificial Analysis with a 10,000-token prompt, in the fastest configuration it measured [5]. Time to first answer token includes the model's reasoning, so a turn takes that time plus its visible output at that speed, plus its requests, which run in parallel within a turn. ferryGL's requests run one after another, at an assumed 0.5 seconds each. 3 models had no measurement, and their times are left out.
- **Schedules and teams**, the same for every model. Two runs a workday (44 a month) or hourly over an eight-hour workday (176 a month), for one user and for teams of 10, 50 and 100, each user with a route of their own.

ferryGL on its own spends no tokens. On a user's machine it costs nothing; in GitHub Actions, each run is a job billed as one minute, at $0.006 a minute for a private repository [25], and a user's 44 runs a month fall well within the 2,000 minutes GitHub Free includes [24]. Table 10 gives the cost without free minutes, as an upper bound.

Results. Tables 9 and 10 and Figure 9 give the results; the data have every combination of model, strategy, assumptions, schedule and team size, per month and per year.

- **At team scale, an agent doing the job itself costs from hundreds to tens of thousands of dollars a year.** For a team of 100 at two runs a workday, it costs $596 to $59,576 a year depending on the model under the plausible assumptions, or $341 to $34,126 under the lean ones. That puts 11.2 to 12.7 billion tokens a year through the model under the plausible assumptions, or 4.8 to 5.8 billion under the lean ones. Hourly runs cost four times as much.
- **The choice of model spans two orders of magnitude.** The dearest flagship costs 100 times as much as the cheapest small model, and within one provider the flagship costs 1.9 to 100 times as much as the cheapest model. Whether a small model applies the rules reliably is what this estimate cannot say.
- **Delegating to ferryGL cuts every model's bill 7 to 21 times**, to $58.99 to $6,249 a year for a team of 100, because the agent reads ferryGL's plan instead of every item.

Table 9. Per run, for one user's route of 100 items, on a run that finds 4 changes to make. Tokens are those of an agent doing the job itself under the plausible assumptions; they differ only by tokenizer. Dollars are for an agent doing the job itself, from the lean to the plausible assumptions, and for one delegating it to ferryGL (plausible). Seconds use the plausible assumptions; — where no speed was measured. [a] Price from 1 December 2026 (Gemini 4 Argon) or 1 January 2027 (Gemini 3.8 Flash); half that until then. [b] Peak-hour price; other hours cost half. [c] Public preview, at its regular price; on sale for half. [d] Released on 19 October 2026.

			Dollars per run		Seconds per run	
Provider	Model	Tokens per run	Doing it itself	Delegating	Itself	Delegating
Anthropic	Claude Fable 5.1	217,857	$0.61–$1.01	$0.11	62	28
	Claude Opus 5.5	217,857	$0.25–$0.42	$0.044	96	50
	Claude Sonnet 5.5	217,857	$0.12–$0.21	$0.022	34	22
	Claude Haiku 5.5	217,857	$0.0065–$0.011	$0.0011	70	44
OpenAI	GPT-6 Astra	217,857	$0.65–$1.13	$0.12	64	28
	GPT-6.1 Sol	217,857	$0.12–$0.21	$0.024	81	31
	GPT-6 Luna	217,857	$0.0065–$0.011	$0.0012	29	22
Google	Gemini 4 Argon[a]	217,857	$0.22–$0.38	$0.050	—	—
	Gemini 3.8 Flash[a]	217,857	$0.088–$0.15	$0.019	164	91
	Gemini 3.5 Flash-Lite	217,857	$0.020–$0.035	$0.0049	54	40
xAI	Grok 4.7	217,857	$0.12–$0.22	$0.015	83	43
	Grok 4.3	217,857	$0.063–$0.11	$0.0071	—	—
DeepSeek	DeepSeek V4 Pro[b]	220,799	$0.065–$0.10	$0.0094	39	25
	DeepSeek V4.1 Flash[b]	220,799	$0.015–$0.025	$0.0026	75	48
Moonshot AI	Kimi K3	212,295	$0.17–$0.29	$0.032	368	175
	Kimi K2.7 Code	212,295	$0.056–$0.10	$0.0091	299	147
Alibaba	Qwen3.8 Max	235,014	$0.12–$0.20	$0.016	415	191
	Qwen3.8 Flash	235,014	$0.0087–$0.014	$0.0012	299	142
Z.ai	GLM-5.3	217,857	$0.080–$0.14	$0.011	171	90
	GLM-5.3-Flash	217,857	$0.0087–$0.016	$0.0012	206	105
MiniMax	MiniMax-M3	212,886	$0.017–$0.032	$0.0028	142	77
	MiniMax-M3.1-Flash[d]	217,857	$0.0079–$0.014	$0.0013	—	—
Mistral AI	Mistral Large 4[c]	217,857	$0.068–$0.12	$0.010	214	108
	Mistral Small 4	240,590	$0.0094–$0.015	$0.0013	25	21
ferryGL alone		0	$0		11	

Table 10. Dollars per year for one user and for teams of 10, 50 and 100, at two runs a workday, under the plausible assumptions (Table 6). A month costs one twelfth of a year; hourly runs cost four times as much. The lean assumptions cost 52 to 63 % as much (Figure 9). ferryGL alone costs nothing on users' own machines, or within GitHub Actions' free minutes; the last row assumes every minute is paid. Notes as in Table 9.

		Agent doing the job itself				Agent delegating to ferryGL			
Provider	Model	1 user	10	50	100	1 user	10	50	100
Anthropic	Claude Fable 5.1	$534	$5,336	$26,682	$53,365	$58.17	$582	$2,908	$5,817
	Claude Opus 5.5	$222	$2,217	$11,087	$22,174	$23.38	$234	$1,169	$2,338
	Claude Sonnet 5.5	$111	$1,109	$5,544	$11,087	$11.69	$117	$584	$1,169
	Claude Haiku 5.5	$5.96	$59.58	$298	$596	$0.59	$5.90	$29.50	$58.99
OpenAI	GPT-6 Astra	$596	$5,958	$29,788	$59,576	$62.49	$625	$3,124	$6,249
	GPT-6.1 Sol	$111	$1,109	$5,544	$11,087	$12.43	$124	$621	$1,243
	GPT-6 Luna	$5.96	$59.58	$298	$596	$0.62	$6.25	$31.24	$62.49
Google	Gemini 4 Argon[a]	$199	$1,993	$9,963	$19,926	$26.53	$265	$1,326	$2,653
	Gemini 3.8 Flash[a]	$80.80	$808	$4,040	$8,080	$9.95	$99.48	$497	$995
	Gemini 3.5 Flash-Lite	$18.62	$186	$931	$1,862	$2.59	$25.88	$129	$259
xAI	Grok 4.7	$118	$1,176	$5,880	$11,760	$8.02	$80.22	$401	$802
	Grok 4.3	$60.51	$605	$3,025	$6,051	$3.77	$37.69	$188	$377
DeepSeek	DeepSeek V4 Pro[b]	$54.64	$546	$2,732	$5,464	$4.98	$49.78	$249	$498
	DeepSeek V4.1 Flash[b]	$12.97	$130	$649	$1,297	$1.40	$13.96	$69.80	$140
Moonshot AI	Kimi K3	$153	$1,530	$7,652	$15,304	$16.91	$169	$846	$1,691
	Kimi K2.7 Code	$53.96	$540	$2,698	$5,396	$4.79	$47.88	$239	$479
Alibaba	Qwen3.8 Max	$104	$1,043	$5,215	$10,429	$8.44	$84.42	$422	$844
	Qwen3.8 Flash	$7.64	$76.38	$382	$764	$0.65	$6.49	$32.45	$64.91
Z.ai	GLM-5.3	$75.45	$755	$3,773	$7,545	$5.69	$56.94	$285	$569
	GLM-5.3-Flash	$8.35	$83.46	$417	$835	$0.64	$6.38	$31.90	$63.80
MiniMax	MiniMax-M3	$16.90	$169	$845	$1,690	$1.46	$14.55	$72.77	$146
	MiniMax-M3.1-Flash[d]	$7.40	$73.98	$370	$740	$0.71	$7.11	$35.56	$71.12
Mistral AI	Mistral Large 4[c]	$63.70	$637	$3,185	$6,370	$5.32	$53.23	$266	$532
	Mistral Small 4	$8.15	$81.47	$407	$815	$0.71	$7.12	$35.61	$71.22
ferryGL alone, own machine		$0	$0	$0	$0				
ferryGL alone, paid Actions minutes		$3.17	$31.68	$158	$317				

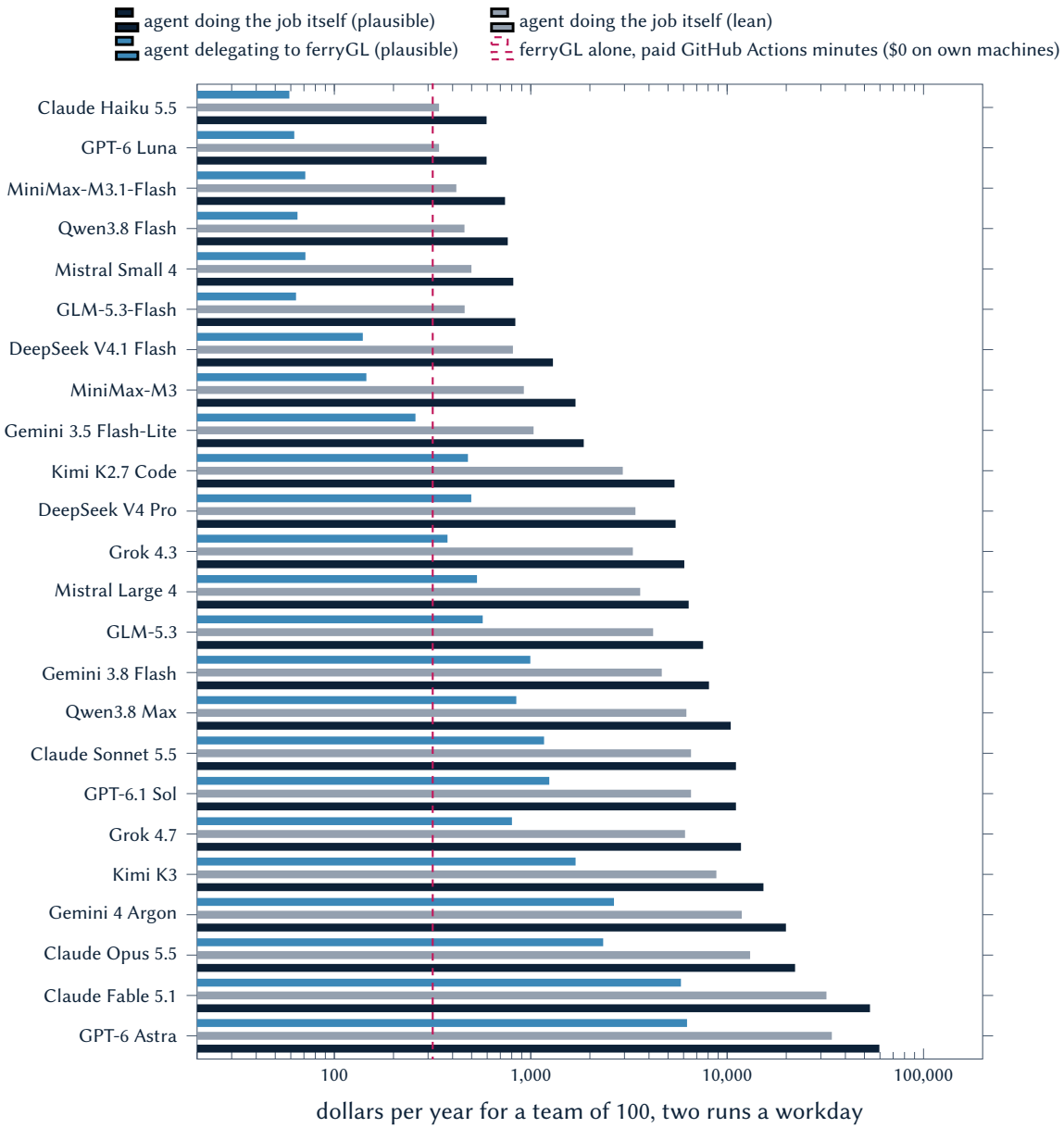


Figure 9. Dollars per year for a team of 100 at two runs a workday, by model, from the cheapest to the most expensive agent. The logarithmic axis spans three orders of magnitude. Delegating to ferryGL cuts every model's bill; ferryGL alone spends nothing on inference, and at most $317 a year on paid GitHub Actions minutes.

Table 11. Ways to track work in another organization's GitHub repositories in your own Linear workspace.

	Linear's GitHub integration	Copying by hand	An AI agent	ferryGL
Needs the other organization's owners or admins	Yes, to install it	No	No	No
Credentials	A GitHub App in their organization	Yours	Yours, given to the agent	Yours: a read-only GitHub token and a Linear API key
Scope	Linked repositories, for the whole team	What you copy	What you ask for	Items you are part of, for you
Cost of each run	None	Your time	Model tokens: 91,620 to 217,857 for 100 items (Section 7.4)	API requests; no model tokens. An agent that runs it: 3,364 to 5,769 tokens
Same result every time	Yes	No	Not guaranteed	Yes
Preview before writing	n/a: it acts on GitHub events	n/a	Depends on the agent	Yes, by default

- **ferryGL alone spends nothing on inference**: $0 on users' own machines or within GitHub Actions' free minutes, and at most $317 a year for a team of 100 on paid minutes. That is less than an agent doing the job itself costs on any model, even under the lean assumptions; delegating to ferryGL costs less than that on the 9 cheapest models, and runs ferryGL as well.
- **ferryGL alone is also the fastest.** A run takes about 11 seconds, nearly all of it its 21 requests to GitHub and Linear at the assumed half second each. An agent doing the job itself takes 25 to 415 seconds, although its requests run in parallel, and a delegating agent, which runs ferryGL twice, 21 to 191 seconds. Models measured only with long reasoning take minutes.

Caveats. Prices change often, and several here are introductory, set by the time of day, or for a preview (notes a to c of Table 9); the script reruns the comparison from a table of prices. Speeds were measured by a third party, on its own prompts, which ask for longer answers than an agent's tool calls. A model measured only with reasoning on is likely slower here than it would be on short tool calls; a model measured without reasoning is timed in a faster configuration than the one whose reasoning tokens are priced, which favors it. Models whose makers publish open weights are counted with the published tokenizer, which the hosted model may not match exactly. Flat-fee subscription plans, and batch discounts, which don't suit an interactive loop, are not modeled.

7.6 Comparison with the alternatives

Table 11 compares ferryGL with the ways people bridge the gap today. The two other automated options differ from ferryGL in what they need from the other organization and in what each run costs. Linear's integration is the right tool when it can be installed: it is maintained by Linear, covers every issue of a repository for the whole team, and can even sync issues both ways [45]. General workflow-automation services offer GitHub and Linear connectors too [48, 54, 79], but the user would have to build the roles, homes, riding and edit-preserving rules as workflows, and some services need an organization owner or administrator to connect a GitHub organization [69].

7.7 Threats to validity

The simulation's conclusions are only as good as its stand-ins. They follow Linear's and GitHub's documented behavior, but real services differ in details the stand-ins leave out, such as the order of results (which the metamorphic check varies), eventual consistency between a write and the next read, and rate limiting. Reads and writes against a real Linear workspace had not yet been exercised at the time of writing; GitHub reads have been exercised against a real repository. The random histories follow hand-chosen event weights, so they cover the rules' cases broadly but do not represent any real person's work. The cost model relies on GitHub's published formula, which GitHub may change. The benchmark ran on one machine. The agent comparison is a cost model built on measured workloads, not a run of an agent; Section 7.4 lists its assumptions.

8 Related work

Synchronization and replication. File synchronizers detect updates by comparing each replica with an archive of the last synchronized state, and propagate the changes [7, 59]; optimistic replication generalizes this, with conflict detection and resolution when both sides change [62, 68], and conflict-free replicated data types avoid conflicts by construction [66]. ferryGL has no conflicts to resolve, because it splits ownership: the GitHub state is authoritative for edges, and the user for everything in between. It keeps the archive in the destination, in link metadata, rather than on the synchronizing machine. Bidirectional transformations, or lenses, address the view-update problem of propagating edits of a view back to its source [23]; ferryGL's Linear issues are a lossy, one-way view, and edits to them never flow back.

Integration and reconciliation. In the vocabulary of enterprise integration patterns [38], ferryGL is a polling consumer and message translator whose receiver is made idempotent by looking up earlier results by key, here the GitHub URL. Helland argues that distributed systems must make such repeated work harmless [37], and HTTP defines which methods are idempotent [22]. A proposed `Idempotency-Key` header would let HTTP clients retry creates safely [40]; Linear's API offers a similar means, an identifier the client chooses for each issue or comment it creates [47]. Kubernetes-style controllers reconcile continuously toward a desired state [10]; ferryGL shows why that level-triggered model is ill-suited when the destination has an owner of its own, and how recording the last acted-on state turns polling into edge-triggered updates.

Issue tracking and pull-based development. Empirical studies describe the issue tracker as a hub of communication and coordination [9], and pull-based development as the dominant way to contribute to projects, including from outside the core team [32–34]. Developers switch often between activities and tools, which they perceive as costly [49]. ferryGL addresses one such switch: keeping a personal tracker in step with work done in someone else's repositories.

AI agents, tools and their cost. Language-model agents that interleave reasoning with tool calls [76] can work in repositories and resolve real GitHub issues [41, 75], and AI assistance can speed up programming [58]. Such an agent could also do ferryGL's job. But the same models are not deterministic in practice, even with settings meant to make them so [6, 57], and they cost tokens on every run. Models can learn to call tools [64], act through executable code [70], and write tools for later use [11], and protocols such as the Model Context Protocol connect agents to them [2]. On cost, cascades that route requests to cheaper models reduce it [13]; evaluations of agents increasingly report it alongside accuracy [42], and a simple, non-agentic pipeline has outperformed open-source software agents on real GitHub issues at low cost [74]; and efficiency has long been argued to be a goal of AI research in its own right [65]. Systems that combine models with conventional components are increasingly the norm [77]. ferryGL takes the view that a routine, rule-based task is better done by a small deterministic program,

token-free, which leaves the agent for work that needs it, and gives the agent a tool to call when the task comes up.

Software citation. This paper, and the `CITATION.cff` file in ferryGL's repository [19], follow the software citation principles [67]: software should be citable, with a persistent identifier and version.

9 Limitations and future work

- **Polling.** Linear catches up only when ferryGL runs. Webhooks would need administrator access in the other organization's repositories. A template for running ferryGL on a schedule with GitHub Actions is planned.
- **Limits of GitHub's API.** A search returns at most 1,000 results, the commit scan reads the first 100 commits of each pull request with up to 5 authors each, and ferryGL reads the first 20 issues GitHub says a pull request will close (mentions in its text are not limited). Review requests to a team the user is on do not count as the user's own.
- **Roles.** Commenting on an item or being mentioned in it does not bring it into scope. Whether either should is an open question.
- **No cleanup.** When an item leaves the user's scope, its Linear issue stays as it is. This is deliberate, since nothing is ever deleted, but it means a Linear issue can outlive the user's involvement.
- **Order dependence.** The case found in Section 7.1 should be fixed, with a planning test that states the scenario, and the order-independence check kept as a regression test.
- **Retrying creates.** A comment whose creation failed in transit may be posted twice. Choosing the identifier of each new issue and comment on the client, which Linear's API allows [47], would let ferryGL retry creates without that risk.
- **Live validation.** The Linear client is tested against Linear's published schema and against fakes, but not yet against a real workspace.
- **A live agent comparison.** The agents' costs are estimated, and their correctness not measured. Running real agents over the same histories would measure both.
- **Agent interfaces.** Agents use ferryGL through its command line and read its text output. Machine-readable output, or a Model Context Protocol server [2], would let agents call it without a shell; neither exists yet.

10 Conclusion

ferryGL brings a person's work in other organizations' GitHub repositories into their own Linear workspace with only their own credentials, and without asking anything of the other organization. Its design keeps the synchronization state where the synchronized data lives, in the metadata of the links between the two, and acts on edges in the GitHub state rather than on its level, which together give idempotent reruns without a database and leave the user's own edits alone. A pure planner, a client interface that cannot express the forbidden writes, and eight explicit invariants make it safe to run in repositories that belong to someone else. A randomized simulation of tens of thousands of runs with failing writes found no violation of those invariants, and found one ordering edge case worth fixing. An AI agent doing the same job itself would, by this paper's estimate, spend 54,966 to 217,857 tokens on every run of a 100-item route, including the runs that find nothing to do; an agent that delegates the job to ferryGL spends a few thousand, growing only slowly with the route. Under the plausible assumptions, across 24 models from 10 providers, that is the difference between $596–$59,576 and $58.99–$6,249 a year for a team of 100, against nothing on inference for ferryGL alone. Both lessons generalize. Any

one-way copy between two systems where the destination has an owner of its own can keep its memory in the destination and act on edges. And any routine, rule-based task that agents are asked to repeat is a candidate for token-free automation: written once as a deterministic tool, it costs API requests instead of inference, behaves the same on every run, and leaves the agent's tokens for the work that needs them.

Availability

ferryGL is free software under the MIT license. The source is at https://github.com/ao-batouche/ferryGL, releases are published on PyPI as `ferrygl`, and the documentation is at https://oussamabatouche.com/ferryGL/. This paper describes version 0.2.1. Its source, the evaluation scripts, and the data behind every table and figure are in the repository, under `docs/paper/`.

Declaration of AI use

AI tools were used to build and maintain ferryGL and to write this paper. AI coding agents were used to build and maintain the software, working under the contributor rules in the repository's `AGENTS.md`, which state the invariants of Section 5. This paper, including its figures, the evaluation scripts and the verification of its references against their sources, was drafted with Anthropic's Claude (Claude Opus 5.5, through Claude Code). The author directed this work, reviewed and edited its output, and takes full responsibility for the software and for the content of this paper. In line with COPE's position, AI tools are not listed as authors [17].

Trademarks

ferryGL is an independent project. It is not affiliated with or endorsed by Linear or GitHub. Linear and GitHub are trademarks of their respective owners.

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A A complete configuration

Only `[[route]]` is required. `comment`, `[titles]`, `[copy]` and `[status]` show their defaults; `project`, `github_user`, `since`, `label` and `api_key_file` are optional, and unset by default.

```
[[route]]
repos = ["acme/api", "acme/web"]
team = "ENG"                  # Linear team key or name
project = "Acme engagement"   # optional
github_user = "alice-acme"    # optional: read GitHub as this gh account

[[route]]
repos = ["upstream-org/*"]    # every repository of a user or organization
team = "ENG"

[sync]
since = 2026-09-01            # ignore items not updated since this date
label = "ferryGL"             # optional: an existing label for issues ferryGL creates
comment = false               # true: comment "(BY-FGL) ..." on each change
# Roles, all true by default: issues_assigned, issues_created, prs_created,
# prs_assigned, prs_reviewed, review_requests, prs_with_commits

[titles]                      # use "{repo}#{number}" to keep titles out of Linear
issue = "{title} #{number}"
pr = "{title} - #{number}"
involved = "[PR] {title} - #{number}"

[copy]
description = false

[status]                      # your team's Linear workflow state names
todo = "Todo"
in_progress = "In Progress"
in_review = "In Review"
done = "Done"
canceled = "Canceled"

[linear]
api_key_file = "~/.config/ferrygl/linear-api-key"
```

B Reproducing the evaluation

From a checkout of the repository, with uv installed:

```
uv sync
uv run python docs/paper/evaluation/simulate.py   # writes data/simulation.*
uv run python docs/paper/evaluation/costs.py      # writes data/github-cost.csv
uv run python docs/paper/evaluation/bench.py      # writes data/planner-time.csv
uv run python docs/paper/evaluation/agent_cost.py # writes data/agent-*
uv run --with tiktoken --with tokenizers --with huggingface_hub \
  python docs/paper/evaluation/count_tokens.py     # writes data/tokenizers.csv
uv run python docs/paper/evaluation/compare_models.py  # writes data/models*
make -C docs/paper                                 # builds the PDF with latexmk
```

The simulation is deterministic for a given seed (20261019 by default, one scenario per consecutive seed), and exits with status 1 while any property fails, as order independence does in this version. The cost

experiment asserts that its request counts match the model and that ferryGL sent no mutation. Planner timings vary with the machine; the figure's data was measured as described in Section 7.3.